

Francesco Moca

Blockchain & Full-Stack Developer · Milan, Italy

francesco.moca@gmail.com github.com/franzmoca linkedin.com/in/francesco-moca francescomoca.ml

SUMMARY

Blockchain and full-stack developer with 5+ years of professional experience delivering enterprise distributed-ledger applications. Deep expertise across the full project lifecycle — technical analysis, smart contract development, Java/Spring Boot backends, Vue.js frontends, and cloud infrastructure. Comfortable working in both delivery and architecture roles, and experienced leading small teams on complex projects.

EXPERIENCE

Blockchain Reply

Mar 2020 – Present

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Designed and delivered 13+ enterprise blockchain applications for clients in financial services, energy, insurance, supply chain, and digital entertainment. Operated in roles spanning full-stack developer, technical lead, and on-site blockchain architect, depending on project phase and scope.

- Wrote smart contracts in **Solidity** (ERC-20, ERC-721, AMM / Uniswap v3 forks, security tokens), **Go chaincode** (Hyperledger Fabric), and **PyTeal** (Algorand ASA-020), deploying across Polygon, Hyperledger Besu, Hyperledger Fabric, R3 Corda, and Algorand mainnet and testnets.
- Built **Java / Spring Boot** backends — REST APIs, event listeners, and integration layers — for token lifecycle management, DeFi exchange operations, and digital asset issuance platforms.
- Developed **Vue.js 2/3** web frontends for institutional and consumer-facing products, including a permissioned DEX dashboard, NFT marketplaces, and energy-asset tokenisation portals.
- Integrated institutional custody solutions — **Gnosis Safe** multisig and **Fireblocks** — enabling segregated vault management for regulated financial products.
- Orchestrated **Hyperledger Fabric** networks on **Kubernetes / AWS EKS**; managed infrastructure-as-code with Terraform and CloudFormation.
- Acted as on-site blockchain architect at a major payments provider, documenting solution architectures and overseeing migration of a production **R3 Corda** network (Italian interbank reconciliation system).
- Led technical analysis phases, authored architecture documents, mentored junior developers, and contributed to pre-sales technical proposals.

Stack: Java 8/11/17 · Spring Boot · Go · TypeScript · Solidity · PyTeal · Vue.js 2/3 · Quasar · Vuetify · PostgreSQL · Keycloak · Kubernetes · AWS · Terraform · Docker · R3 Corda · Hyperledger Fabric/Besu · Algorand · Polygon · Gnosis Safe · Fireblocks

Freelance Developer

Sep 2016 – Feb 2020

Blockchain & Mobile Developer · Remote

Delivered client projects alongside university studies, including an ERC-721 smart contract system with a React web dashboard and React Native mobile app for NFT-based product authenticity (Ethereum), and a cross-platform board game companion app published on the iOS App Store and Google Play, maintained through multiple releases.

Stack: Solidity · React · React Native · Ionic · AngularJS · TypeScript · Ethereum

SKILLS

Languages	Java · Go · TypeScript · JavaScript · Solidity · PyTeal
Backend	Spring Boot · Hibernate · Node.js · Keycloak
Frontend	Vue.js 2/3 · React · React Native · Quasar · Vuetify · Ionic
Blockchain	Ethereum / EVM · Algorand · Hyperledger Fabric · Hyperledger Besu · R3 Corda · Polygon · Gnosis Safe · Fireblocks
Infrastructure	Kubernetes · AWS (ECS, EKS) · Terraform · CloudFormation · Jenkins · Ansible · Docker
Data	PostgreSQL

EDUCATION

Master's Degree in Computer Science (Software Techniques)

2016 – 2019

University of Bologna · **110/110 con lode**

Thesis: *Pig — A State Channel Game on Ethereum*. Implemented the dice game Pig as a trustless Layer 2 application using counterfactual instantiation: players exchange signed state updates off-chain and settle only the final outcome on-chain, reducing blockchain interactions to two transactions per game regardless of length.

Bachelor's Degree in Computer Science

2013 – 2016

University of Perugia · **105/110**

Thesis: *Blockchain: gestione di Big Data*. Analysis of the Bitcoin blockchain as a large-scale distributed data structure, with an interactive force-directed visualisation of transaction graphs built in D3.js to explore value-flow patterns between addresses.

CERTIFICATIONS & LANGUAGES

Professional Scrum Developer I (PSD I)

Scrum.org · 2022

Ethereum Developer Program Bootcamp

ConsenSys Academy · 2019

Italian

Native

English

Professional — advanced written, intermediate spoken